
Environmental exposure and child health risk

Bihar — district-level ML diagnostics

Figures package

Generated 2026-05-02

Model summary

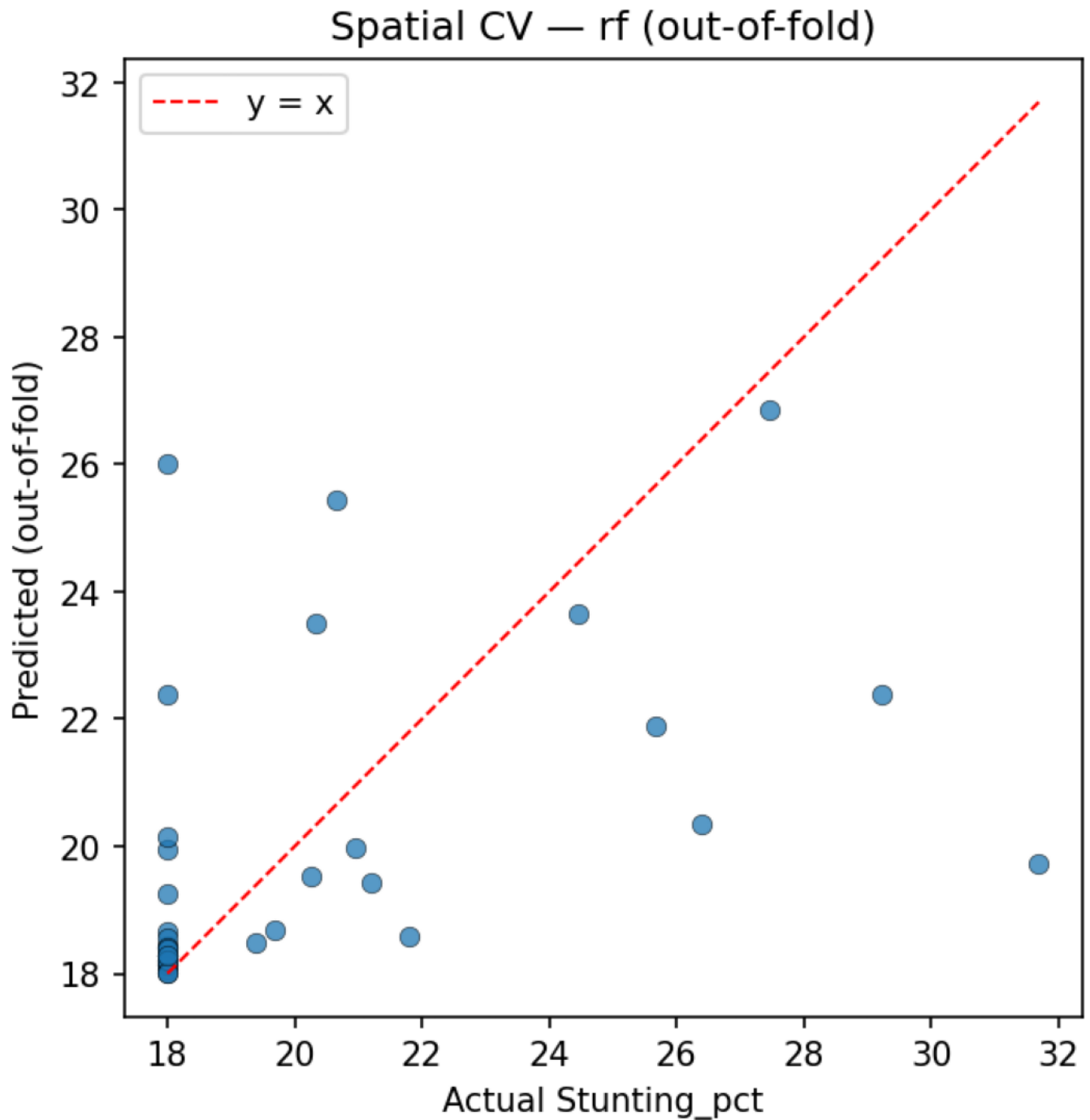
Districts / rows: 38 | CV folds: 5
Selected model: RF

Cross-validation (spatial, by district):

- ridge: RMSE 3.079 | MAE 2.337 | R^2 -0.100
- rf: RMSE 2.961 | MAE 1.822 | R^2 -0.007
- xgb: RMSE 3.180 | MAE 1.929 | R^2 -0.185

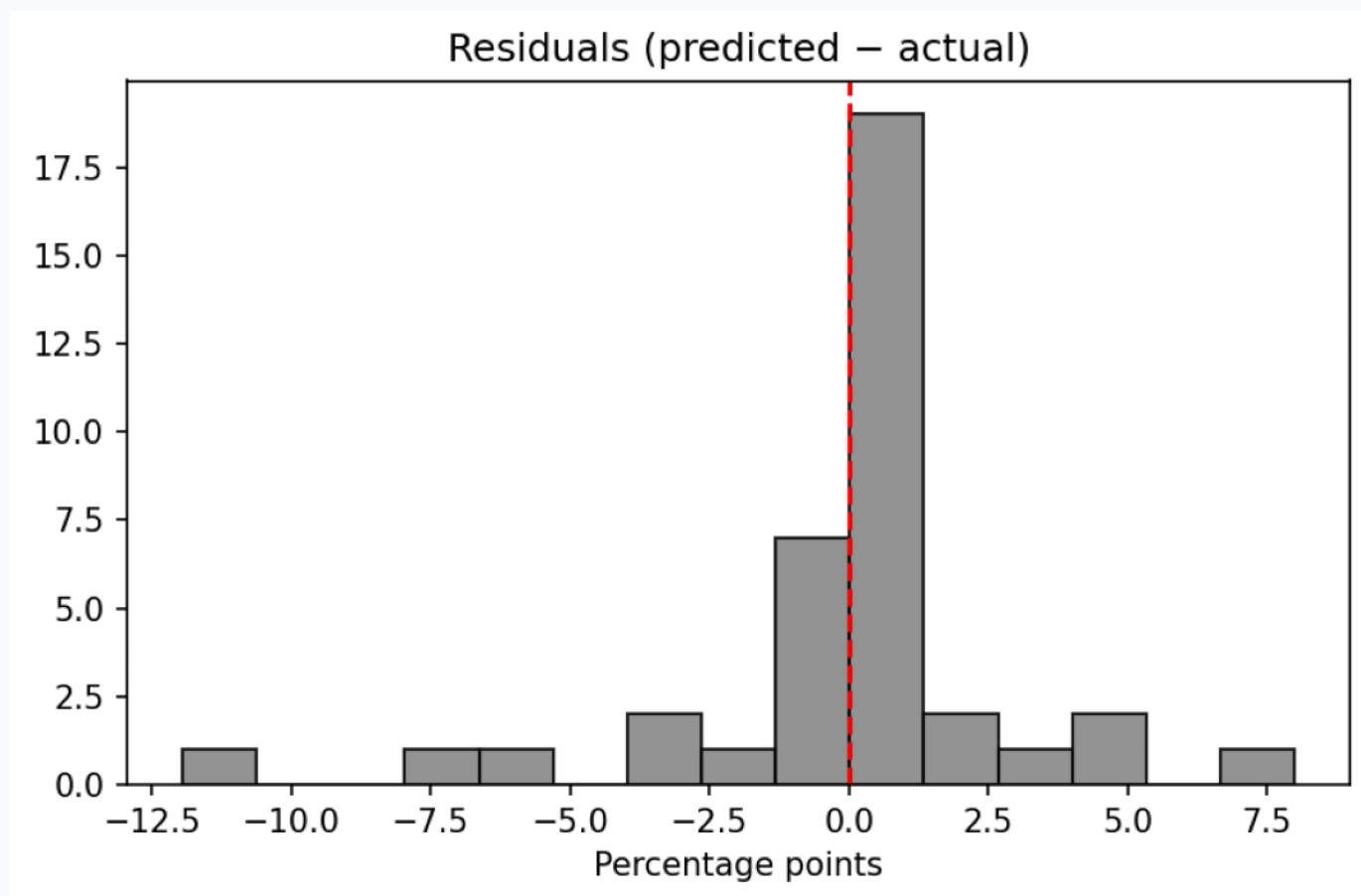
Note: Current demo training data may be synthetic. Replace with official merged extracts (NFHS, CGWB, rainfall, Census) before policy use.

Actual vs predicted (spatial CV)



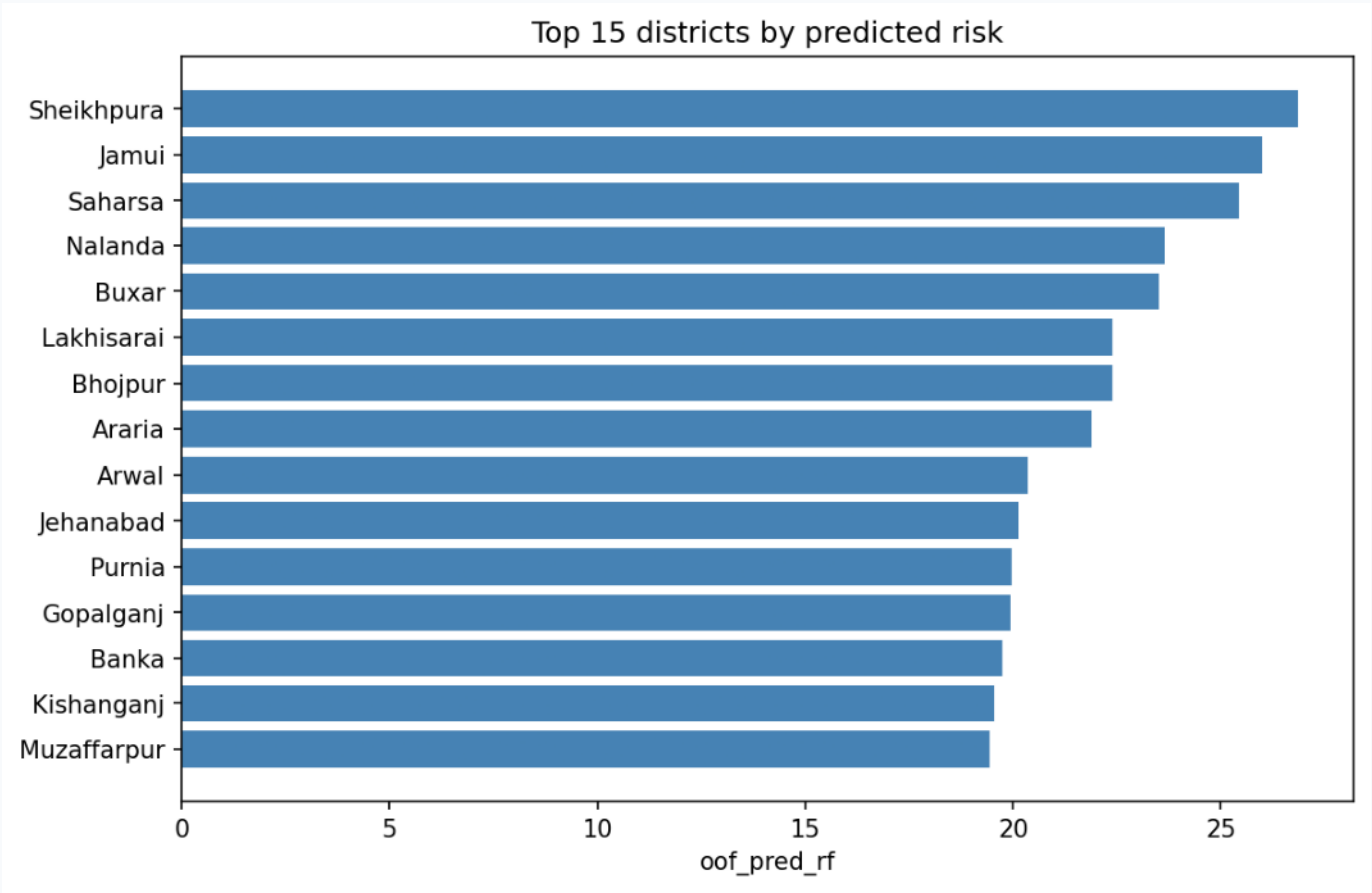
Each point is one district; predictions are out-of-fold under GroupKFold by district. The dashed line is perfect agreement ($y = x$).

Residual distribution



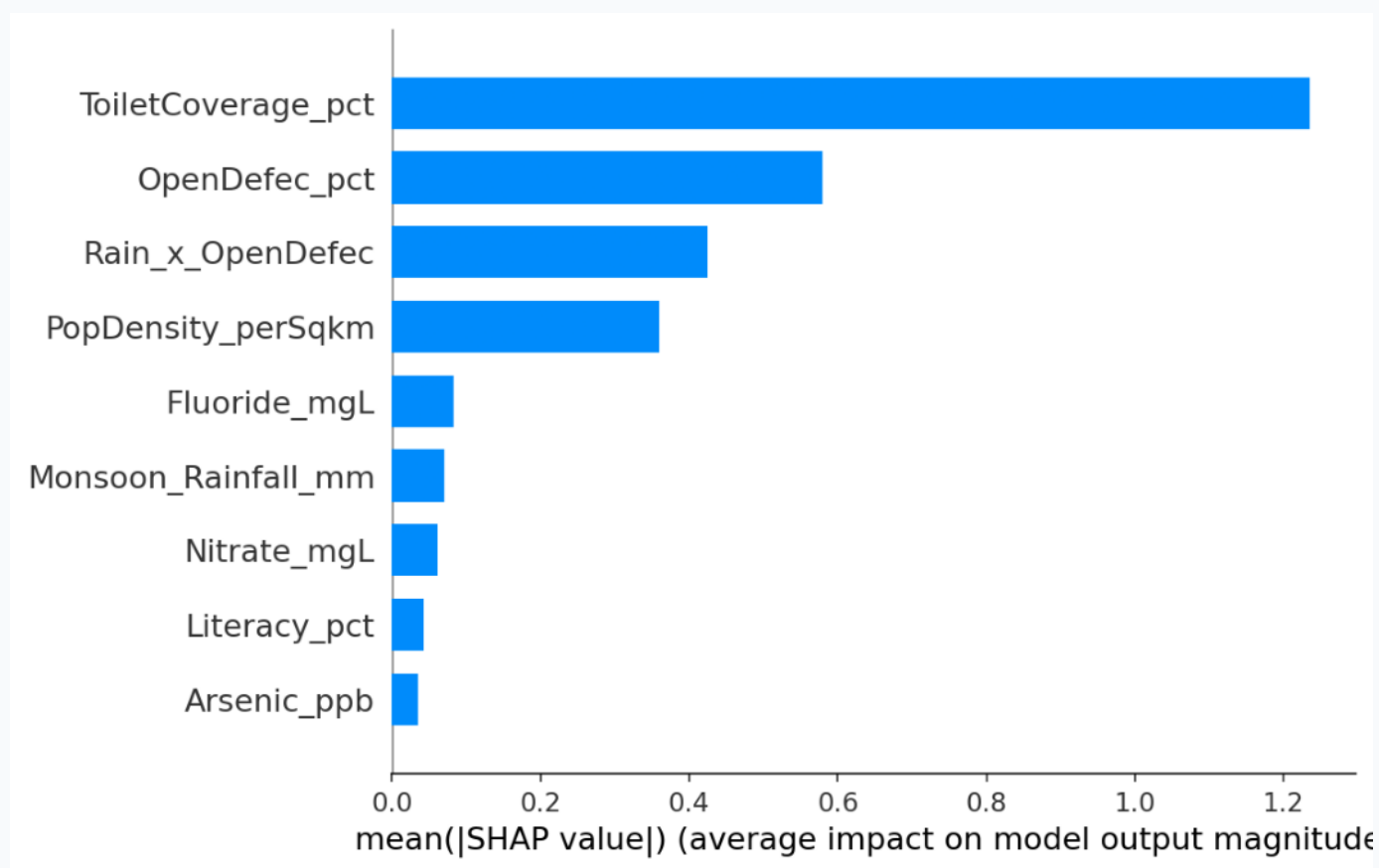
Histogram of prediction errors (predicted – actual) in percentage points.

Highest predicted risk districts



Districts ranked by out-of-fold predicted stunting prevalence for the selected model.

SHAP global importance



Mean absolute SHAP value per feature for the fitted best model (full-data refit).